

CONTACT INFORMATION	Phone: +81-08063217094 E-mail: xinqiang.cai@riken.jp , jksrdivide@gmail.com
BIOGRAPHY	[Google Scholar, dblp] Currently I am a postdoctoral researcher in the Imperfect Information Learning Team at RIKEN Center for Advanced Intelligence Project, led by professor Masashi Sugiyama.
EDUCATION	The University of Tokyo, Tokyo, Japan Ph.D., Complexity Science and Engineering, Oct., 2021 – Sep., 2024 • Laboratory: Sugiyama-Yokoya-Ishida Lab, led by Professor Masashi Sugiyama • Advisor: Professor Masashi Sugiyama • Research Area: Reinforcement Learning and Imitation Learning from Weak Supervision Nanjing University, Nanjing, China M.S., Computer Science and Technology, Sep., 2018 – Jun., 2021 • Laboratory: LAMDA Lab, led by Professor Zhi-Hua Zhou • Advisor: Co-supervised by Professor Zhi-Hua Zhou and Professor Yuan Jiang • Research Area: Machine Learning and Data Mining Northwestern Polytechnical University, Xi'an, China B.E., Aircraft Design and Engineering, Sep., 2014 – Jun., 2018
RESEARCH INTERESTS	Reinforcement Learning, Imitation Learning, Learning with Weak Supervision, LLM Post-Training
PUBLICATIONS (* DENOTES EQUAL CONTRIBUTIONS)	[1] Zhiqiang Kou*, Junyang Chen*, Xin-Qiang Cai, Xiaobo Xia, Ming-Kun Xie, Dong-Dong Wu, Biao Liu, Yuheng Jia, Xin Geng, Masashi Sugiyama, Tat-Seng Chua. Positive-Unlabeled Reinforcement Learning Distillation for On-Premise Small Models. In: Proceedings of the 43rd International Conference on Machine Learning (ICML'26), Seoul, South Korea, Jul. 6-11, 2026. [2] Soichiro Nishimori, Xin-Qiang Cai, Johannes Ackermann, Masashi Sugiyama. Leveraging Domain-Unlabeled Data in Offline Reinforcement Learning across Two Domains. In: Proceedings of the second Reinforcement Learning Conference (RLC'25), Edmonton, Aug. 5-9, 2025. [3] Jing-Cheng Pang, Nan Tang, Kaiyuan Li, Yuting Tang, Xin-Qiang Cai, Zhen-Yu Zhang, Gang Niu, Masashi Sugiyama, Yang Yu. Learning View-invariant World Models for Visual Robotic Manipulation. In: Proceedings of the Thirteenth International Conference on Learning Representations (ICLR'25), Singapore, Apr. 24-28, 2025. [4] Zelei Cheng, Xian Wu, Jiahao Yu, Shuo Han, Xin-Qiang Cai, Xinyu Xing. Soft-Label Integration for Robust Toxicity Classification. In: Proceedings of the Thirty-eighth Conference on Neural Information Processing Systems (NeurIPS'24), Vancouver, Canada, Dec. 10-15, 2024. [5] Yuting Tang*, Xin-Qiang Cai*, Yao-Xiang Ding, Qiyu Wu, Guoqing Liu, Masashi Sugiyama. Reinforcement Learning from Bagged Reward. Transactions on Machine Learning Research (TMLR). [6] Xingyu Song, Zhan Li, Shi Chen, Xin-Qiang Cai, Kazuyuki Demachi. An Animation-based Augmentation Approach for Action Recognition from Discontinuous Video. In: Proceedings of the 27th European Conference on Artificial Intelligence (ECAI'24), Santiago de Compostela, Oct. 19-24, 2024.

- [7] Kaiyan Zhao, Qiyu Wu, **Xin-Qiang Cai**, Yoshimasa Tsuruoka. Leveraging Multi-lingual Positive Instances in Contrastive Learning to Improve Sentence Embedding. In: **Proceedings of the 8th Conference of the European Chapter of the Association for Computational Linguistics (EACL’24)**, Malta, Mar. 17-22, 2024.
- [8] Pushi Zhang*, Baiting Zhu*, **Xin-Qiang Cai***, Li Zhao, Masashi Sugiyama, Jiang Bian. IG-Net: Image-Goal Network for Offline Visual Navigation on A Large-Scale Game Map. In: **Proceedings of the Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS’23)**, 6th Robot Learning Workshop, New Orleans, US, Dec. 10-16, 2023.
- [9] **Xin-Qiang Cai**, Pushi Zhang, Li Zhao, Jiang Bian, Masashi Sugiyama, Ashley Juan Llorens. Distributional Pareto-Optimal Multi-Objective Reinforcement Learning. In: **Proceedings of the Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS’23)**, New Orleans, US, 2023.
- [10] **Xin-Qiang Cai**, Yu-Jie Zhang, Chao-Kai Chiang, Masashi Sugiyama. Imitation Learning from Vague Feedback. In: **Proceedings of the Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS’23)**, New Orleans, US, 2023.
- [11] **Xin-Qiang Cai**, Yao-Xiang Ding, Zi-Xuan Chen, Yuan Jiang, Masashi Sugiyama, Zhi-Hua Zhou. Seeing Differently, Acting Similarly: Heterogeneously Observable Imitation Learning. In: **Proceedings of the Eleventh International Conference on Learning Representations (ICLR’23, spotlight)**, Kigali, Rwanda, 2023.
- [12] Zi-Xuan Chen*, **Xin-Qiang Cai***, Yuan Jiang, Zhi-Hua Zhou. Anomaly Guided Policy Learning from Imperfect Demonstrations. In: **Proceedings of the 21th International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS’22, oral)**, Auckland, New Zealand, 2022.
- [13] **Xin-Qiang Cai**, Yao-Xiang Ding, Yuan Jiang, and Zhi-Hua Zhou. Imitation Learning from Pixel-Level Demonstrations by HashReward. In: **Proceedings of the 20th International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS’21, oral)**, online, 2021.
- [14] **Xin-Qiang Cai**, Peng Zhao, Kai Ming Ting, Xin Mu, Yuan Jiang. Nearest Neighbor Ensembles: An Effective Method for Difficult Problems in Streaming Classification with Emerging New Classes. In: **Proceedings of the 19th IEEE International Conference on Data Mining (ICDM’19)**, Beijing, China, 2019.

PREPRINTS

- [1] **Xin-Qiang Cai**, Masashi Sugiyama. VI-CuRL: Stabilizing Verifier-Independent RL Reasoning via Confidence-Guided Variance Reduction. [arXiv](#).
- [2] Zhiqiang Kou, Junyang Chen, **Xin-Qiang Cai**, Ming-Kun Xie, Biao Liu, Changwei Wang, Lei Feng, Yuheng Jia, Gang Niu, Masashi Sugiyama, Xin Geng. Rethinking Toxicity Evaluation in Large Language Models: A Multi-Label Perspective. [arXiv](#).
- [3] **Xin-Qiang Cai**, Wei Wang, Feng Liu, Tongliang Liu, Gang Niu, Masashi Sugiyama. Reinforcement Learning with Verifiable yet Noisy Rewards under Imperfect Verifiers. [arXiv](#).
- [4] Jiansong Wan, Chengming Zhou, Jinkua Liu, Xiangge Huang, Xiaoyu Chen, Xiaohan Yi, Qisen Yang, Baiting Zhu, **Xin-Qiang Cai**, Lixing Liu, Rushuai Yang, Chuheng Zhang, Sherif Abdelfattah, Hayong Shin, Pushi Zhang, Li Zhao, Jiang Bian. PIG-Nav: Key Insights for Pre-trained Image Goal Navigation Models. [arXiv](#).
- [5] Zelei Cheng*, **Xin-Qiang Cai***, Yuting Tang, Pushi Zhang, Boming Yang, Masashi Sugiyama, Xingyu Xing. UC-MOA: Utility-Conditioned Multi-Objective Alignment for Distributional Pareto-Optimality. [arXiv](#).
- [6] Yuting Tang*, **Xin-Qiang Cai***, Jing-Cheng Pang, Qiyu Wu, Yao-Xiang Ding, Masashi Sugiyama. Beyond Simple Sum of Delayed Rewards: Non-Markovian Reward Modeling for Reinforcement Learning. [arXiv](#).

WORKING
EXPERIENCE

Part-Time

- Microsoft Research Asia (MSRA), Intern, Dec. 2023 to Mar. 2024.

Full-Time

- RIKEN Center for Advanced Intelligence Project (AIP), Postdoc, Oct. 2024 to now.

TEACHING
EXPERIENCE

- [1] 2026 S1S2 (Graduate). Special Topics in Mechano-Informatics II, The University of Tokyo. Guest lecture: “An Introduction to Reinforcement Learning”.
- [2] 2026 Spring (Undergraduate/Graduate). Special Lecture on Information Science IV, Ochanomizu University. Guest lecture: “An Introduction to Reinforcement Learning”.

FUNDINGS

- 2026–2028 JSPS KAKENHI – Grant-in-Aid for Early-Career Scientists, PI (5,000,000 JPY)
- 2024–2026 JSPS KAKENHI – Grant-in-Aid for JSPS Fellows, PI (1,500,000 JPY)
- 2022 Microsoft Research Asia Collaborative Research Program Fellowship (10,000 USD)
- 2021 SPRING GX Fellowship (1,000,000 JPY)

HONORS AND
AWARDS

Contest Awards

- 2018 First Prize, ZhongAn Hackathon Contest, AI scenario
- 2017 Silver Medal, ACM/ICPC Asia Regional Contest, Qingdao Site
- 2017 Bronze Medal, ACM/ICPC Asia Regional Contest, Xinjiang Site
- 2016, 2017 First Prize, China Collegiate Programming Contest
- 2016, 2017 Honorable Mention, Interdisciplinary Contest In Modeling
- 2016 Bronze Medal, ACM/ICPC Asia Regional Contest, Dalian Site
- 2016 Bronze Medal, ACM/ICPC Asia Regional Contest, Shenyang Site
- 2016 Bronze Medal, ACM/ICPC EC-Final Contest, Shanghai Site
- 2015 Second Prize, China Undergraduate Mathematical Contest in Modeling

Honors

- 2024 Star of Tomorrow Award, Microsoft Research Asia (MSRA)
- 2024 JSPS DC2 Fellowship, The University of Tokyo
- 2021 SPRING GX Fellowship, The University of Tokyo
- 2021 Excellent Graduate of Nanjing University
- 2020 Excellent Graduate Student of Nanjing University
- 2020 AAMAS 2021 Attendance Scholarship
- 2019 ICDM Student Travel Award
- 2018, 2019, 2020 First Prize, Nanjing University Academic Scholarship
- 2018 Second Prize, Nanjing University Yingcai Scholarship
- 2018 Excellent Graduate Student of Northwestern Polytechnical University
- 2017 First Prize, Wu Yajun Alumni Scholarship
- 2015, 2016 National Scholarship

SERVICES

Conference Research Services

- Reviewer for ICML, NeurIPS, ICLR, ECAI, CIKM, IJCAI, CCML

Journal Research Services

- Reviewer for [IEEE Transactions on Neural Networks and Learning Systems \(TNNLS\)](#), [Transactions on Machine Learning Research \(TMLR\)](#), [Machine Learning](#)

Volunteer Work

- 2021 AAMAS (Proceedings of the 20th International Conference on Autonomous Agents and Multi-Agent Systems)
- 2018, 2019 MLA (China Symposium on Machine Learning and Applications)

COMPUTER SKILLS	Programming: Python, C/C++, Matlab, Java, Shell, $\text{\LaTeX}$ Operating systems: Ubuntu/Linux, Windows family Languages: Chinese (Native), English (TOEFL-iBT 105/120), Japanese (Intermediate)
HOBBIES	Fitness, Hiking, Running